Teaching notes

This resource is designed to help pupils learn to answer extended response questions effectively.

Students have the word list in front of them plus various coloured pencils.

Use the PPT to reveal the three clues one at a time. After each clue is revealed, students decide which words to 'save or scrap', before showing the next clue. Then show students the exam question.

They are then asked to underline certain words using different colours. At this point go over some of their ideas to iron out any misconceptions.

When this is done students use prompts to help them plan an answer the question. A simple mark scheme is revealed and students write an answer to the question.

As an alternative to using the PPT the clues, instructions and mark scheme could be printed on paper and handed out to the students at the appropriate time.

Clue 1

The question is about plants.

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Clue 2

The question is linked to how substances are carried around a plant.

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Clue 3

The question only concerns the substances involved in the production of glucose.

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The question

How are the reactants and products of photosynthesis transported around a plant?

6 marks

Instructions

- underline the substances in blue
- underline the structured involves in transport in red
- underline the processes involved in green.

Look out for words which might mean the same thing.
E.g. glucose and dissolved sugars.

Planning your answer

Now you are ready to plan your answer the question

Write short statements which link a structure to a substance.

E.g. 'Root hair cells absorb water'

Decide which statement has the strongest link to transport in plants (remember the question is asking about the transport of substances around the plant so place more emphasis on this rather then how substances enter or leave a plant – although this is still relevant).

Put your statements in order starting with this one first and ending with the weakest statement.

Link each statement to a process

E.g. 'Root hair cells absorb water by osmosis'

Are there any other words remaining in your 'save or scrap' list which are relevant? Plan how you will use these.

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Writing your answer

You are now fully prepared. Use your statements, your 'save or scrap' list and the mark scheme to write a full answer to the question.

Mark scheme

Level 1 1 – 2 marks: The answer includes one substance OR one named structure OR one process connected to the transport system in plants.

Level 2 3 – 4 marks: A simple explanation which links a substance to a named structure and it's transportation process OR two structures to their correct processes OR two substances each linked to a process involved in their transportation.

Level 3 5 – 6 marks: A detailed explanation that links water to xylem and glucose to phloem and the processes involved and including at least one other process involved in the transport of substances.

Keywords

haemoglobin	phototropism	photosynthesis	dissolved sugars	transpiration
anthers	leaves	light	pollen	mosquito
kidney	starch	stomata	carbon dioxide	oxygen
xylem	phloem	water vapour	stamen	root hair cells
concentration gradient	evaporation	diffusion	veins	vascular bundle
translocation	transpirational pull (stream)	neutralisation	osmosis	mineral ions
glucose	fungi	peristalsis	ovary	stomach acid

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